

Department of CSE [Data Science]

Cross-Channel Marketing Funnel Leakage Intelligence

TAE-II Winter 2026 (2023 Pattern)

Business Intelligence(23UDSPEL4704A)—B.Tech Final Year(Sem-VII)

Guided By:

Mr. Chinmay Mukim

Name of Student:-

Aman Yadav , Roll No. : P-07

Ghananshu Mali , Roll No. : P-16

Contents

- Introduction
- Objectives & Conversion Funnel
- End-to-End Business Intelligence Architecture
- Multi-Source Data Ingestion & Automated ETL
- Data Cleansing & 34-Point Data Quality Engine
- Dimensional Data Warehouse & Star Schema
- Dashboard Insights — Executive Overview & Funnel Leakage
- Customer Insights & Business Recommendations
- Conclusion

Introduction

Modern digital marketing generates customer interactions across multiple platforms such as Google, Meta, Email and CRM systems. However, these interactions often remain fragmented across different data sources.

Key Problems

- Data Fragmentation: Marketing data is distributed across multiple platforms and systems.
- Visibility Gap: Continuous customer journeys from impression → purchase are difficult to reconstruct.
- Funnel Leakage: Users may click advertisements but drop off before product discovery, cart or checkout.
- Budget Inefficiency: Channels that generate traffic may not necessarily generate conversions or revenue.
- Decision-Making Gap: Marketing managers need a unified Business Intelligence view to identify where and why customers are being lost.

Core Problem Statement

How can fragmented cross-channel marketing data be transformed into actionable Business Intelligence for identifying funnel leakage, evaluating channel performance, and optimizing marketing decisions?

Objectives & Conversion Funnel

➤ Project Objectives

1. Data Integration
2. Ingest and clean 57,000+ multi-touch customer journey interactions.
3. Data Warehousing
4. Construct an enterprise Star Schema warehouse using MySQL with conformed dimensions.
5. Automated ETL
6. Develop a modular Python ETL pipeline with automated validation and zero-silent-drop validation.
7. Business Intelligence
8. Develop an interactive 4-page Power BI dashboard with reconciled DAX metrics

➤ 7-Stage Conversion Funnel

Impression → Click → Landing Page → Product View → Cart → Checkout → Purchase

➤ Funnel Concept

For each stage:

Stage Drop-off = Users entering previous stage – Users progressing to next stage

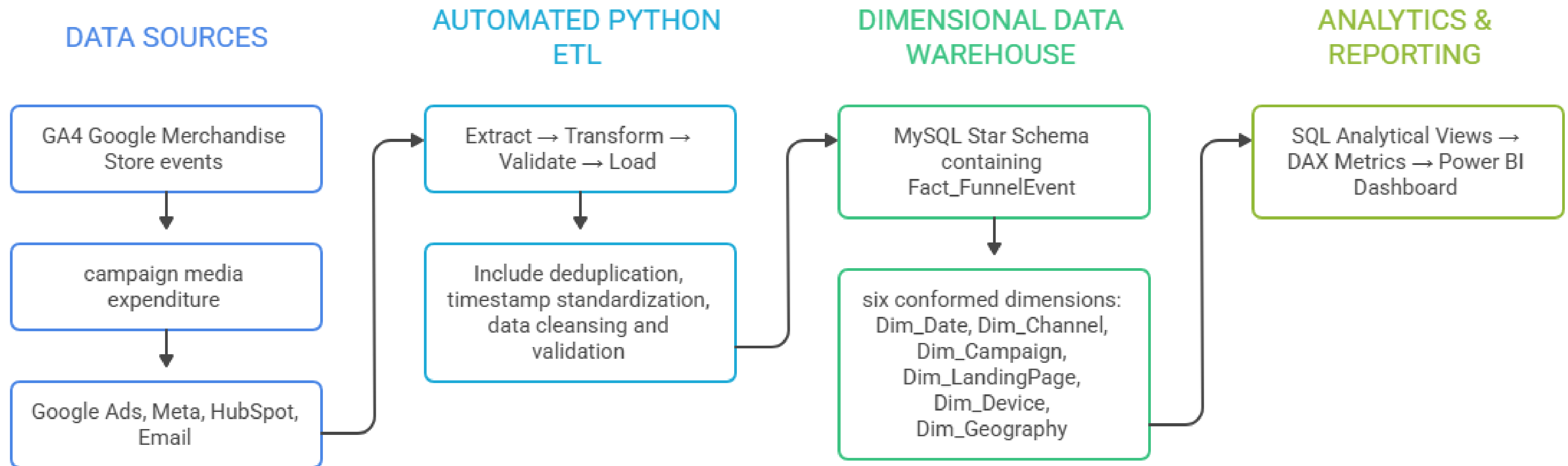
Stage Conversion Rate = Users progressing to next stage / Users entering previous stage × 100

➤ Key Purpose

Identify where customers leave the journey and quantify the business impact of that leakage.

End-to-End Business Intelligence Architecture

Create a 4-layer Business Intelligence architecture for a Cross-Channel Marketing Funnel Leakage Intelligence system

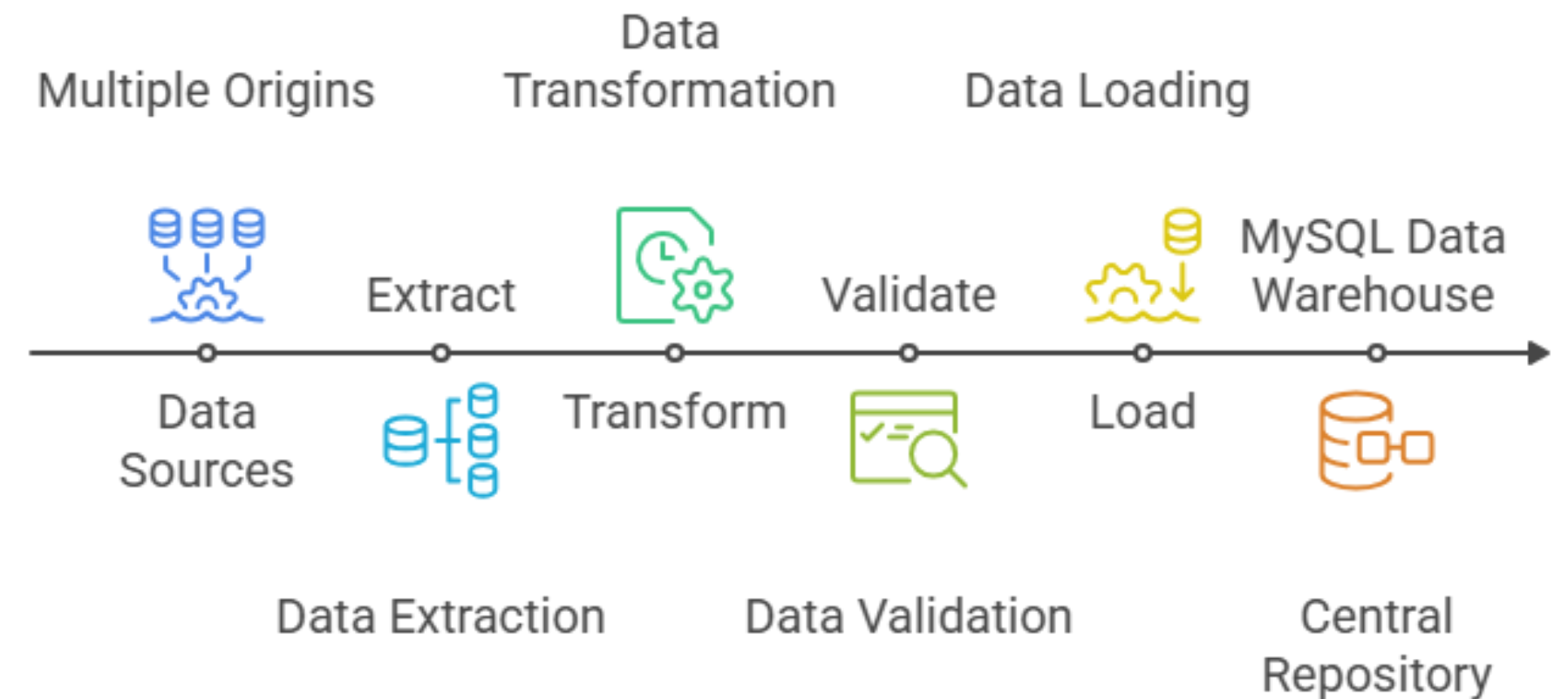


Multi-Source Data Ingestion & Automated ETL

Data Sources

- 1. GA4**
 - Customer behaviour
 - Events
 - Sessions
 - Journey progression
- 2. Campaign Spend**
 - Daily media expenditure
 - Campaign-level spending
- 3. External Marketing Platforms**
 - Google Ads
 - Meta
 - HubSpot
 - Email

Modular Python ETL Pipeline for Business Intelligence: -



Data Cleansing & 34-Point Data Quality Engine

✦ Data Cleansing

- Flatten nested GA4 schemas.
- Convert Unix microsecond timestamps into standardized datetime values.
- Remove orphan events.
- Normalize traffic-source and media values.
- Map traffic media into conformed channel categories.
- Apply proportional ad-spend allocation.
- Prevent downstream double-counting.

✦ Validation Result

34 / 34 Checks Passed

100% Validation Pass Rate

✦ Data Quality Validation

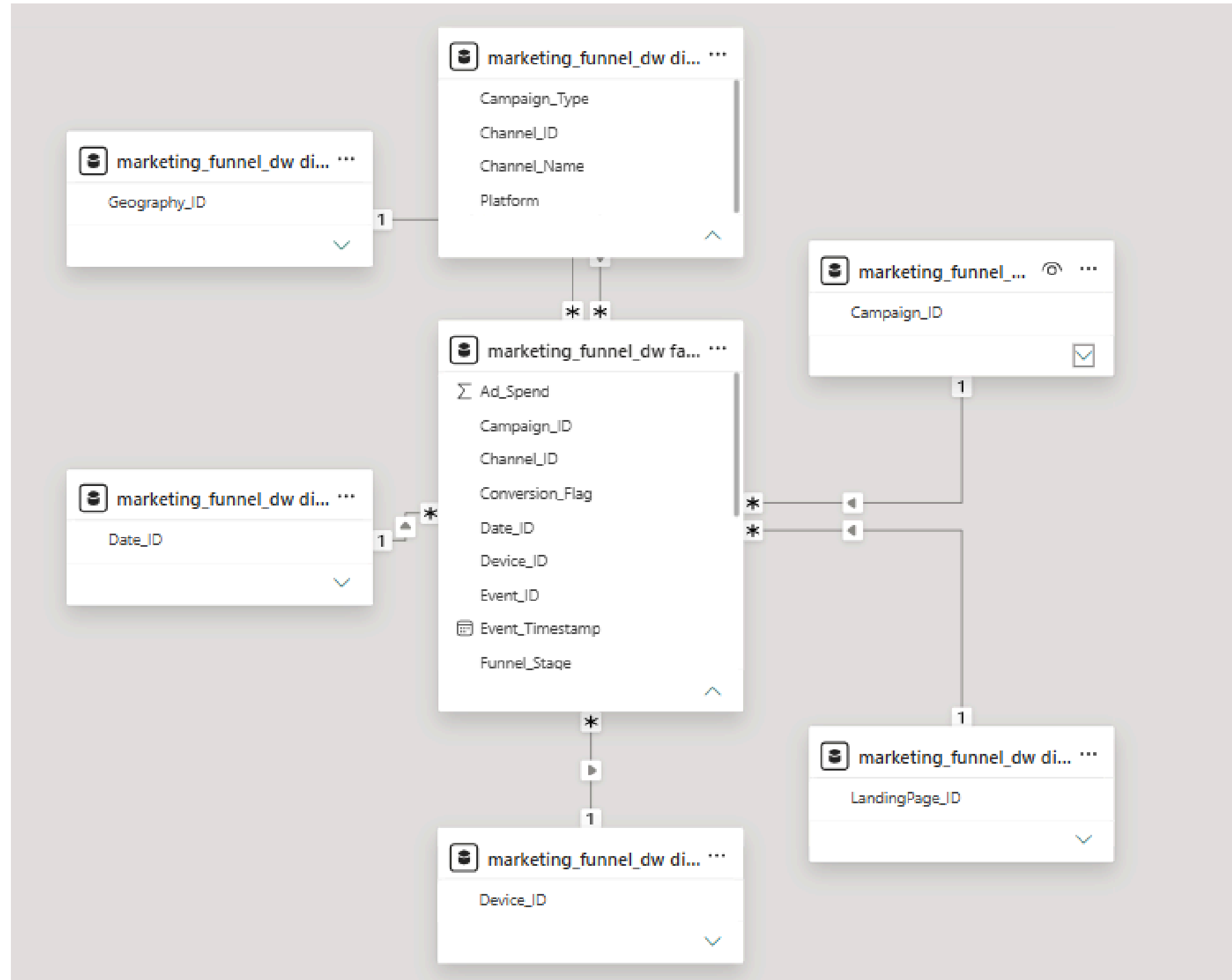
The automated validation engine checks:

- Required columns
- Primary-key uniqueness
- Foreign-key referential integrity
- Stage validity
- Duplicate records
- Data consistency
- Schema conformity

✦ Key Benefit

Ensures that only validated and structurally consistent data reaches the analytical warehouse.

Dimensional Data Warehouse & Star Schema



Warehouse

Database: MySQL

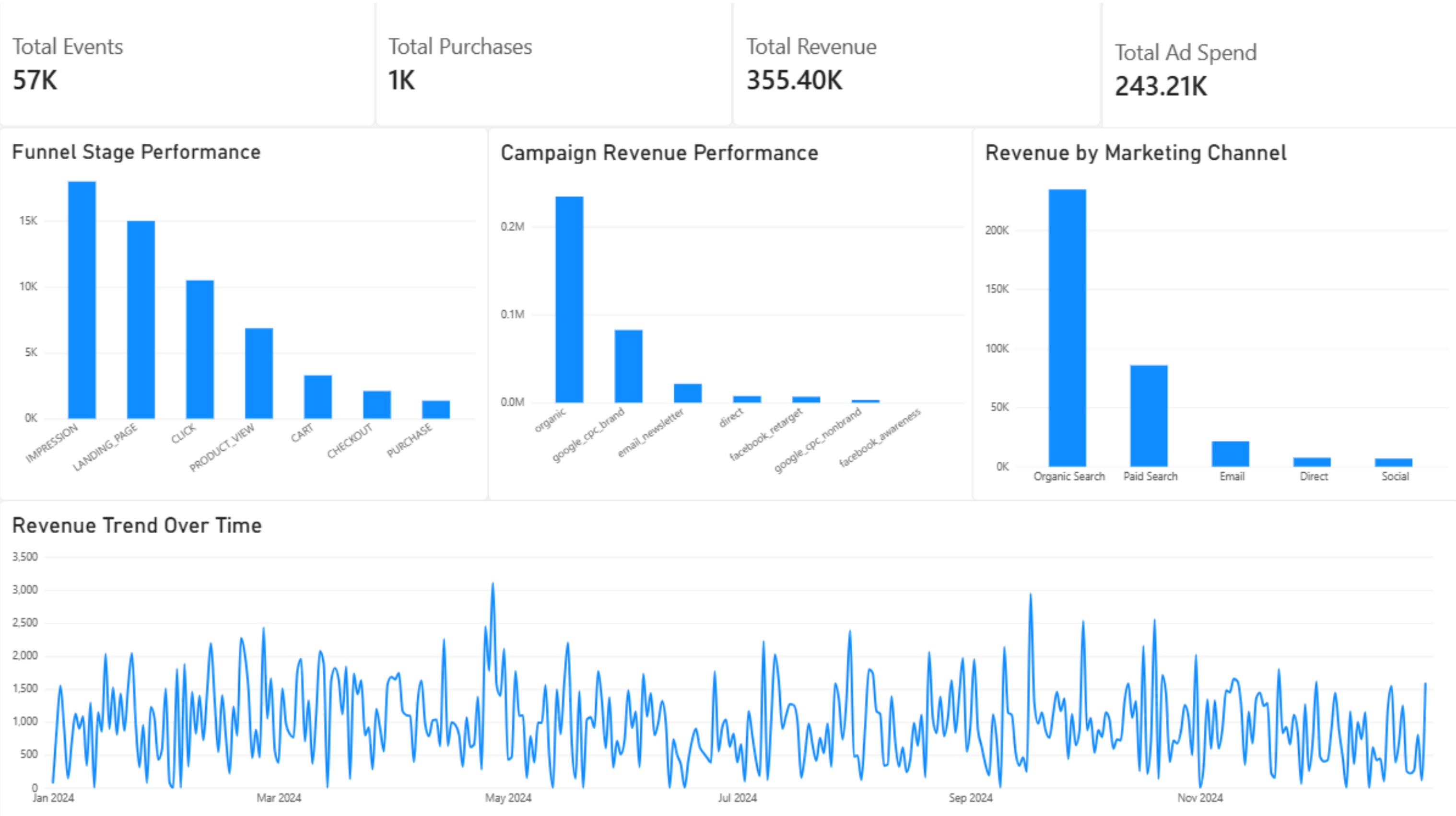
Warehouse Benefits

- Consistent analytical structure
- Faster querying
- Reusable dimensions
- Reduced data redundancy
- Consistent reporting across channels

Performance

Sub-second aggregation
across 57,000+ events

Dashboard Insights – Executive Overview & Funnel Leakage



Customer Insights & Business Recommendations

Customer / Device Insight

Mobile = 56.4% of Sessions

However → Mobile Checkout Abandonment = 42.1%

This indicates a significant difference between mobile traffic volume and completed purchase behaviour.

Other Segmentation Areas

- Geographic performance distribution
- Landing-page conversion efficiency
- Device-level behaviour
- Customer journey stage performance

Business Recommendations

1. Reallocate Awareness Budget

Reallocate approximately 15–20% of awareness budget toward:

- High-intent retargeting
- Cart-recovery sequences

2. Improve Mobile Checkout

Introduce a simplified checkout experience, including:

- One-click digital wallets
- Reduced friction
- Faster completion flow

3. Improve Landing Pages

Improve:

- Product discoverability
- Product recommendation widgets
- Landing-page journey continuity

Conclusion

The proposed Business Intelligence solution transforms fragmented cross-channel marketing data into a unified analytical framework for customer journey analysis. By integrating GA4, campaign expenditure, and other marketing data sources, the system provides a consistent view of customer interactions. The automated Python ETL pipeline cleans, validates, and prepares the data for reliable analysis. A MySQL Star Schema provides a structured foundation for scalable analytical queries. SQL views and reconciled DAX measures enable consistent KPI and funnel calculations. Power BI then converts these processed datasets into interactive and actionable insights. The analysis identifies significant leakage across the customer conversion journey, particularly at the Landing Page to Product View stage. Channel, campaign, device, and demographic analysis further reveal differences in marketing performance and customer behaviour. These insights can support informed decisions regarding budget allocation, landing-page improvements, and mobile checkout optimization. Overall, the solution demonstrates how Business Intelligence can convert complex marketing data into actionable funnel intelligence and support data-driven decision-making.

Final Takeaway

From fragmented marketing data to actionable funnel intelligence.

Thank you !